



Microstructure Evolution and shear strength optimization in SiC ceramic/MA956 ODS steel joints brazed with Cu-Ti-Si filler



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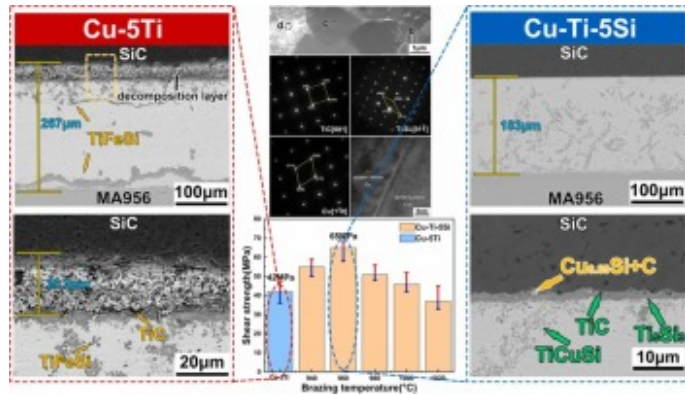
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Abstract

Brazing of SiC ceramic is of importance for its application, but the decomposition of SiC is harmful to the performance of SiC brazed joint. In our work, Cu-Ti-Si filler was designed to braze SiC ceramic and MA956 ODS steel. The effects of brazing temperature and holding time on the microstructure evolution and mechanical properties of the joints were investigated, and the microstructure evolution mechanism was explored. The joint fabricated at 960 °C for 10 min exhibited a typical interfacial microstructure consisting of SiC / Cu_{6,69}Si + graphite / TiC / Ti₅Si₃ / TiCuSi + Ti₅Si₃ + Cu(s,s) / TiCuSi + Cu(s,s) / TiFeSi + Cu(s,s) / MA956. Continuous layered decomposition products from SiC were exclusively observed at the maximum brazing temperature of 1020°C, compared to the Cu-5Ti filler without Si addition, a significant suppression of SiC decomposition was achieved, while holding

time exhibited negligible influence on SiC decomposition extent. Both elevated brazing temperature and prolonged holding time significantly reduced the braze seam width. The shear strength of the joints originally increased and then decreased with rising brazing temperature. The maximum shear strength of 65 MPa was achieved at 960°C for 10 min, representing a 54.8% enhancement compared to joints fabricated using Cu-5Ti fillers.

Graphical Abstract



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Keywords

Silicon Carbide; MA956 ODS steel; Cu-Ti-Si brazing; Microstructure evolution; Shear strength

1. Introduction

Silicon carbide (SiC) ceramics exhibit superior high-temperature stability, exceptional corrosion resistance, and outstanding irradiation resistance, making them ideal structural materials for advanced nuclear reactor applications. However, their high brittleness and poor machinability pose significant challenges in fabricating complex components [1], [2], [3]. Consequently, joining SiC to metallic materials to form composites has emerged as an effective strategy to advance their practical utilization. Nevertheless, the decomposition of SiC ceramics under elevated temperatures critically limits their performance [4], [5]. Therefore, suppressing SiC decomposition has become a key research focus for its joining.

The Fe-Cr-Al-based oxide dispersion strengthened (ODS) alloy MA956, reinforced with Y_2O_3 nanoparticles uniformly dispersed via mechanical alloying, demonstrates outstanding high-temperature performance in nuclear applications [6], [7], [8]. The pinning effect of these Y_2O_3 effectively inhibits dislocation motion, giving the alloy with superior creep resistance and oxidation stability. This attributes position MA956 as a promising candidate for fourth-generation nuclear reactor systems [9], [10]. Therefore, achieving reliable joining between MA956 ODS steel and SiC ceramics is imperative to fully harness the complementary advantages of both materials, propelling their application in nuclear structural materials as an inevitable trend.

Metal-ceramic joining encompasses various techniques, such as diffusion bonding, transient liquid phase (TLP) bonding, and vacuum brazing [11], [12], [13], [14], [15], [16]. Among these, vacuum brazing is widely recognized as the most suitable method for joining ceramics and metals on account of its convenience and efficiency [17], [18], [19], [20], [21]. To date, research on brazed metal-ceramic joints predominantly employs Ni-based, Ag-based, and Cu-based filler alloys [22], [23], [24], [25], [26], [27]. Wang *et al.* employed a Ni-Mo filler to braze SiC ceramics, achieving well-bonded joints at brazing temperatures of 1300–1350 °C. However, the infiltration of Ni into the SiC ceramic during brazing induced decomposition of the SiC, leading to significant degradation of joint integrity [28]. Song *et al.* successfully achieved the brazing of SiC ceramics using graphene nanoplatelet-reinforced AgCuTi filler. Although the joint strength reached relatively high levels, the high cost of AgCuTi filler and the limited high-temperature serviceability of the joints significantly restricted their practical application scope. In contrast to Ni-based and Ag-based fillers, Cu-based filler alloys offer advantages such as lower cost, superior high-temperature performance, and minimal infiltration into the SiC ceramic, thereby avoiding extensive decomposition of the SiC ceramic [29]. Su *et al.* successfully brazed SiC ceramics to MA956 ODS steel using Cu-5Ti filler [30]. However, the melting temperature of the Cu-5Ti filler reached 1037.4 °C. During the brazing process, the decomposition of SiC ceramics generated a significant amount of aggregated brittle Cu-Si compounds and a low-strength graphite phase. These phases created weak areas, severely degrading the joint's mechanical properties. Notably, adjusting process parameters failed to effectively mitigate the decomposition of SiC ceramics. The significant CTE mismatch between SiC ceramics and MA956 ODS steel substrates induced large residual stresses during cooling in

SiC/MA956 brazed joints. This stress concentration reduced the overall joint strength. Consequently, reducing the brazing temperature was imperative to mitigate thermally induced residual stress and inhibit SiC decomposition [31]. To solve these problems, the study investigates methods to reduce the melting point of Cu-5Ti filler [32]. Based on the decomposition mechanism of Cu-induced SiC at elevated temperatures ($\text{Cu} + \text{SiC} \rightarrow \text{Cu-Si (liquid)} + \text{C}$), enhancing the Si content in the Cu-5Ti filler can thermodynamically suppress the forward reaction, thereby effectively inhibiting SiC decomposition [33].

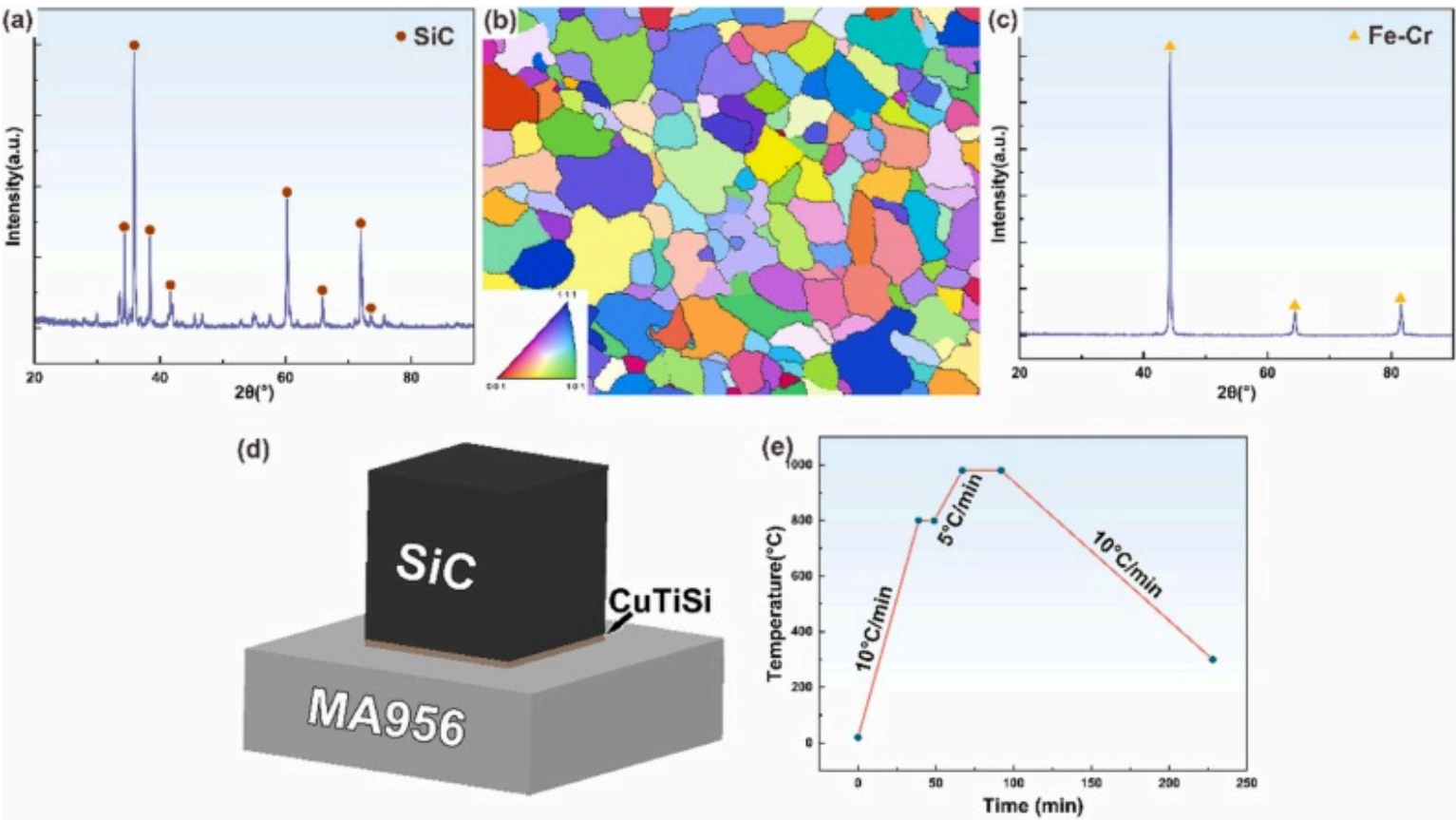


Si element as a common melting point-reducing element, exhibits eutectic reactions with both Cu and Ti. Especially, the coefficient of thermal expansion (CTE) of Si is as low as $2.36 \times 10^{-6} \text{ K}^{-1}$. When incorporated into Cu-Ti filler as a melting point-reducing element, Si not only serves to reduce the melting point of the Cu-Ti filler and minimize the decomposition of SiC ceramics but also helps to alleviate residual stress in the joint to a certain extent.

In this work, the optimal Si content in Cu-5Ti filler was systematically determined by incorporating varying Si additions, effectively suppressing thermal decomposition at the SiC interface through controlled interfacial reactions. The mechanical properties and microstructural evolution of the joints were analyzed to evaluate their viability for nuclear energy applications, and its feasibility was evaluated to facilitate its future application in the nuclear energy field.

2. Materials and methods

The MA956 oxide dispersion strengthened (ODS) steel used in this study was commercially obtained from Shanghai Shanghu Nonferrous Metals Co., Ltd., fabricated via powder metallurgy. The SiC ceramics, supplied by Dongguan Kezhong Ceramics Co., Ltd., were prepared via pressure less sintering with a purity of 98 wt%. The Cu-Ti-Si filler brazing filler material, with a thickness of 200 μm , was synthesized using an arc melting technique. The morphology and X-ray diffraction (XRD) patterns of both MA956 ODS steel and SiC ceramic are presented in Fig. 1(a)-(c), while the chemical composition of the MA956 ODS steel is summarized in Table 1.



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Fig. 1. Base materials characterization and shear testing of joints: (a) XRD pattern of SiC; and (b) EBSD result and XRD pattern (c) of MA956; (d) schematic of brazing assembly; (e) heating curve during brazing.

Table 1. Chemical composition of MA956 ODS steel (wt%).

Element	C	Mn	Y ₂ O ₃	Cr	Ni	Cu	Ti	Co	Al	Fe
Max	-	-	-	18.5	-	-	0.2	-	3.75	Balance

Element	C	Mn	Y ₂ O ₃	Cr	Ni	Cu	Ti	Co	Al	Fe
Min	0.1	0.3	0.5	21.5	0.5	0.15	0.6	0.3	5.75	

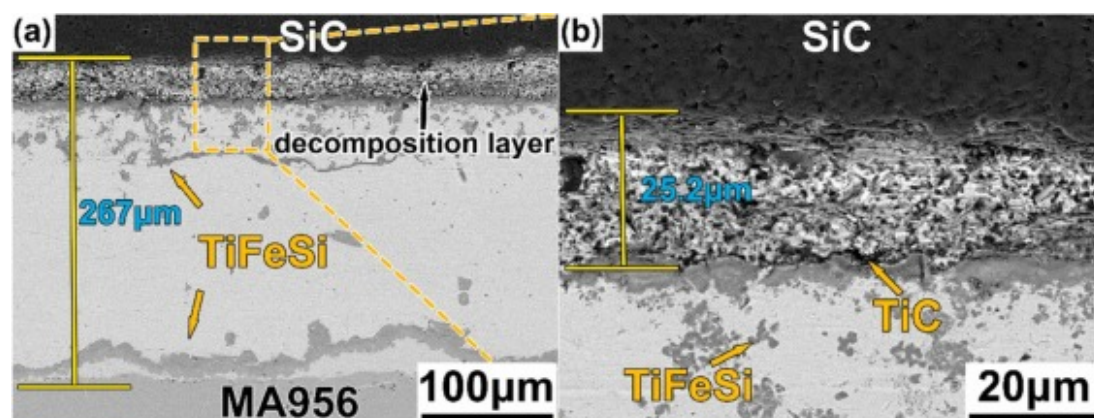
Prior to brazing, MA956 ODS steel and SiC base materials were machined into specimens with dimensions of 10 mm × 10 mm × 3 mm for steel and 5 mm × 5 mm × 5 mm for SiC. The bonding surfaces were sequentially ground with 180-, 400-, and 800-grit papers to remove contaminants and oxide layers, followed by ultrasonic cleaning in ethanol. The Cu-Ti-Si filler foil (200 μm thick) was cut into 5 mm × 5 mm × 5 mm dimensions and similarly cleaned. The components were assembled in a layered configuration with the sequence SiC/Cu-Ti-Si filler/MA956 steel. The assembled specimens were loaded into a VF-1600–222 vacuum brazing furnace. After achieving a base vacuum of 5×10^{-3} Pa, the brazing cycle was initiated: Heating from ambient to 800 °C at 20 °C/min, followed by a 20-min hold; Further heating to the target brazing temperature (940–1020 °C) at 10 °C/min with isothermal retention; Controlled cooling to 300 °C at 5 °C/min, concluding with furnace cooling to room temperature. After brazing completion, specimens requiring microstructural analysis were embedded using dental acrylic powder, sectioned perpendicular to the bonding interface using a SYJ-200 precision cutter, subsequently, the specimens were successively abraded with 180-, 400-, 800-, 1200-, and 2000-grit sandpapers, followed by a polishing treatment[34]. The brazed joints were analyzed by various microscopic techniques, including scanning electron microscopy (SEM; MERLIN Compact, Zeiss), energy dispersive spectroscopy (EDS; OCTANE PLUS, EDAX) and transmission electron microscopy (TEM), for determining the interfacial microstructures and fracture surfaces. Shear tests were conducted at a speed of 0.5 mm/min using an INSTRON 5967 testing machine at room temperature.

3. Result

3.1. Influence of Si content on the microstructure and properties of Cu-Ti-Si filler

Fig. 2 showed the interfacial morphology and local magnification analysis of SiC/Cu-5Ti/MA956 ODS steel brazed joint (T = 1000 °C, t = 10 min). For the Cu-5Ti filler without Si addition (the melting point is 1037.4 °C), intensified interfacial reactions with the SiC substrate occurred at 1100 °C, creating a 25.2 μm thick SiC decomposition layer. This occurred as excessively high brazing temperature promotes Cu-induced accelerated decomposition of SiC ceramics, creating brittle Cu-Si intermetallic and graphite

phases. These phases together increase the thickness of the SiC decomposition layer. Concurrently, elevated brazing temperature amplified residual stress at the joint interface due to the CTE mismatch between SiC ceramics and MA956 ODS steel.



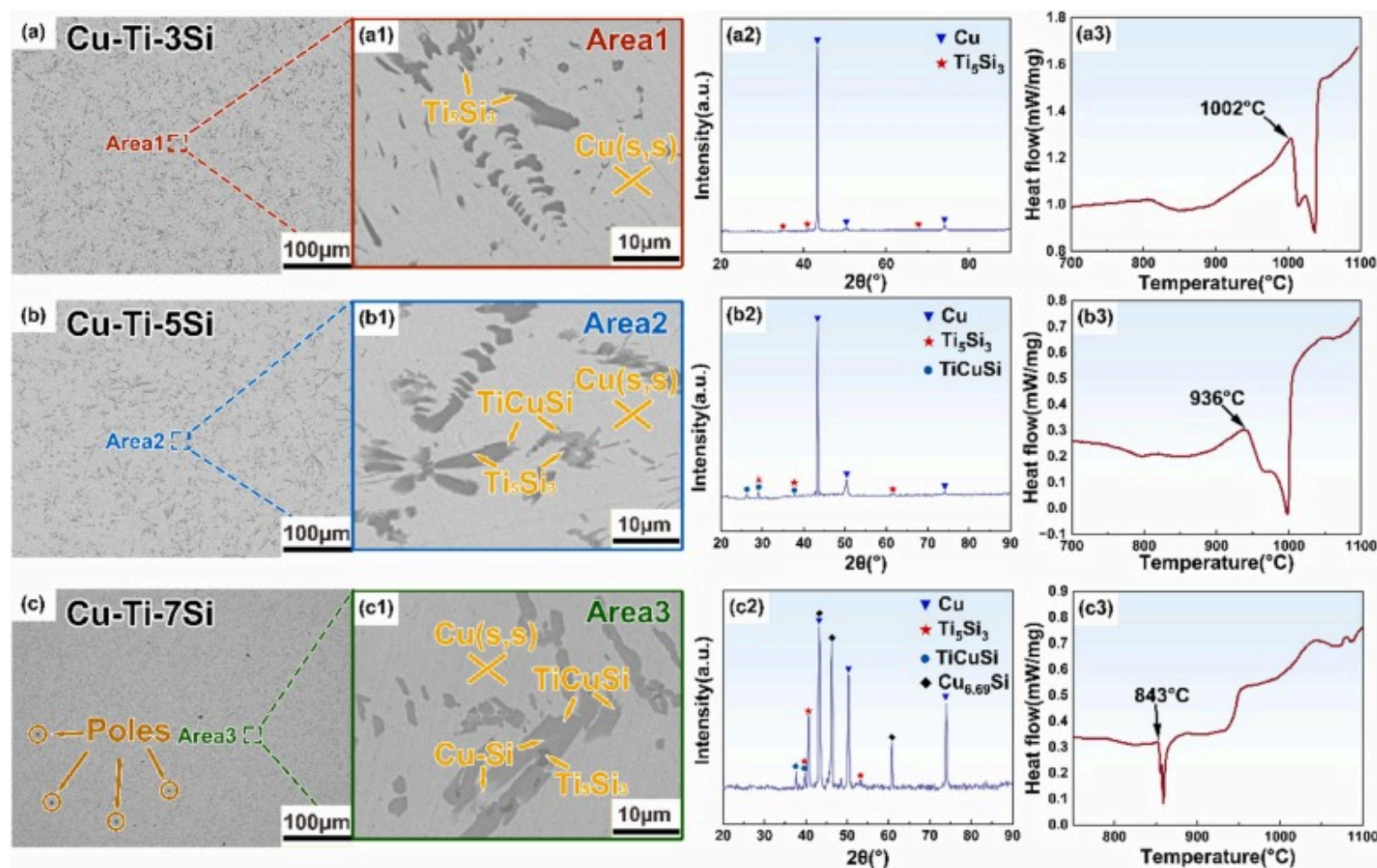
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Fig. 2. Microstructural analysis of the typical interface in the SiC/Cu-5Ti/MA956 ODS steel joint ($T = 1000\text{ }^{\circ}\text{C}$, $t = 10\text{ min}$). (a) Overall morphology; (b) Magnified view of interface.

The microstructure evolution and thermal property characterization of Cu-Ti-Si filler with Si element additions to the Cu-5Ti filler were illustrated in Fig. 3. Su et al. [30] investigated the microstructure and properties of a Si-free Cu-5Ti brazing filler. Their study revealed that the filler's microstructure comprises a Cu-based solid solution and the Cu_4Ti intermetallic compound, with a melting point of 1005°C . Energy-dispersive spectroscopy (EDS) analysis revealed that the 3 wt% Si-containing filler consisted of a light-gray Cu(s,s) and a gray-black Ti_5Si_3 intermetallic compound, exhibiting a uniformly distributed microstructure. The microstructure of the 5 wt% Si-containing filler consisted of three distinct phases: a light gray Cu(s,s), a gray-black Ti_5Si_3 intermetallic compound, and a dark gray TiCuSi ternary phase, with uniform distribution. Compared to the 3 wt% Si filler, the 5 wt% Si filler exhibited an additional TiCuSi phase in its microstructure. The microstructure of the 7 wt% Si-containing filler exhibited a significant increase of micropores. These microvoids led to pronounced embrittlement and a marked decline in the ductility of the filler. These microvoids significantly enhanced brittleness and severely degraded the plasticity of the filler. EDS analysis revealed that the 7 wt% Si-containing filler had a multiphase microstructure with four distinct phases: a light gray Cu(s,s), a gray-black Ti_5Si_3 intermetallic compound, a dark-gray TiCuSi, and off-white Cu-Si phases. XRD results confirmed the

presence of these phases across all compositions. Differential scanning calorimetry (DSC) analysis demonstrated the melting behavior of the filler. Adding 3 wt% Si to the Cu-5Ti filler resulted in a melting point of approximately 1002 °C, with a broader melting range. However, compared to the as-received Cu-5Ti filler, the Si-doped showed only a negligible decrease in melting point, indicating that Si had limited efficiency in lowering the melting point at this composition. Adding 5 wt% Si to the Cu-5Ti filler lowered the melting point to 936 °C, a reduction of approximately 70 °C compared to the as-received Cu-5Ti filler. This demonstrated the effectiveness of Si in reducing the melting point in this system. The 7 wt% Si-modified filler displayed a further lowered melting onset (~843 °C); however, excessive brittle intermetallics (Ti_5Si_3 , TiCuSi) and porosity drastically impaired its ductility and flowability, the brazing system showed poor compatibility for joining base metal with significant CTE mismatch.



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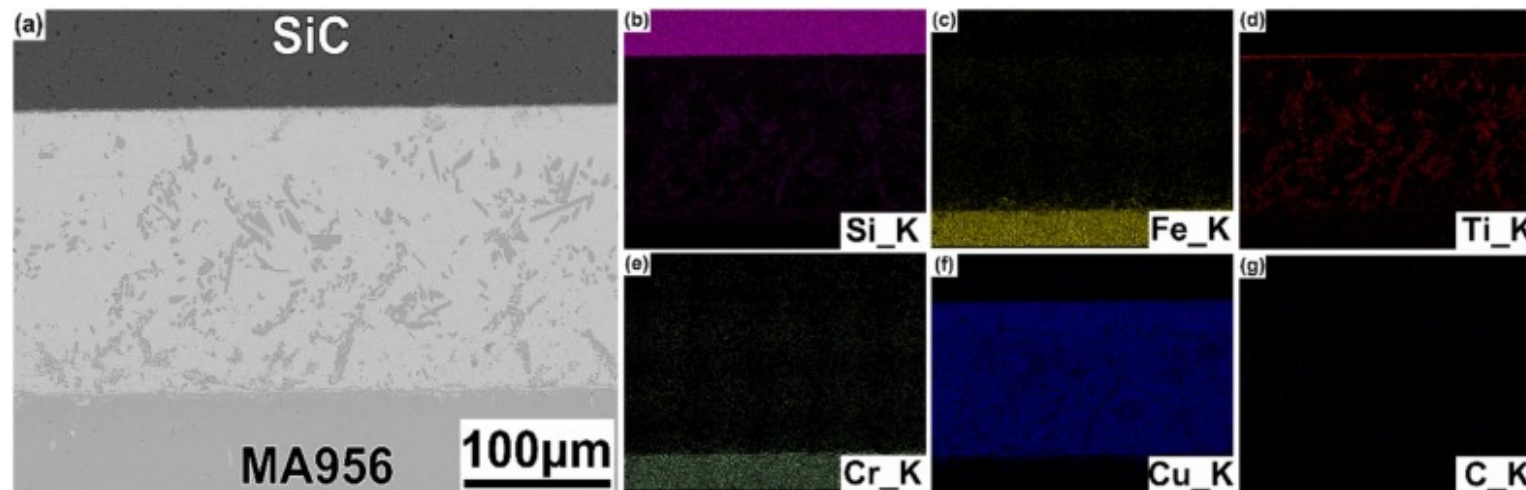
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Fig. 3. Characterization of Cu-Ti-Si filler with varying Si contents:(a-c) SEM macroscopic image of the filler; (a1-c1) SEM localized image of the filler;(a2-c2) XRD patterns. (a3-c3) DSC curve.

Based on the comprehensive analysis above, the Cu-Ti-Si filler with 5 wt% Si addition to the Cu-5Ti system was selected for brazing SiC ceramics to MA956 ODS steel.

3.2. Typical interfacial microstructure of SiC /Cu-Ti-5Si /MA956 brazed joint

Fig. 4 displayed the interfacial microstructure and elemental distribution maps of the SiC/Cu-Ti-5Si filler/MA956 ODS steel joint brazed at 960 °C for 10 min. As observed in Fig. 4(a), the joint exhibited no apparent detectable cracks or voids. Fig. 4(b)–(g) presented the elemental distribution maps of Si, Fe, Ti, Cr, Cu and C. The results indicated that Ti, Cu, and Si were uniformly distributed across the braze seam, while Cr and C showed no significant aggregation. The absence of a carbide interlayer on the SiC side indicated that SiC did not decompose significantly during the brazing process. The SiC side was primarily composed of Ti, Si, and Cu elements. The MA956 side showed additional Fe enrichment, indicating significant Fe dissolution and diffusion from MA956 ODS steel at the brazing temperature.



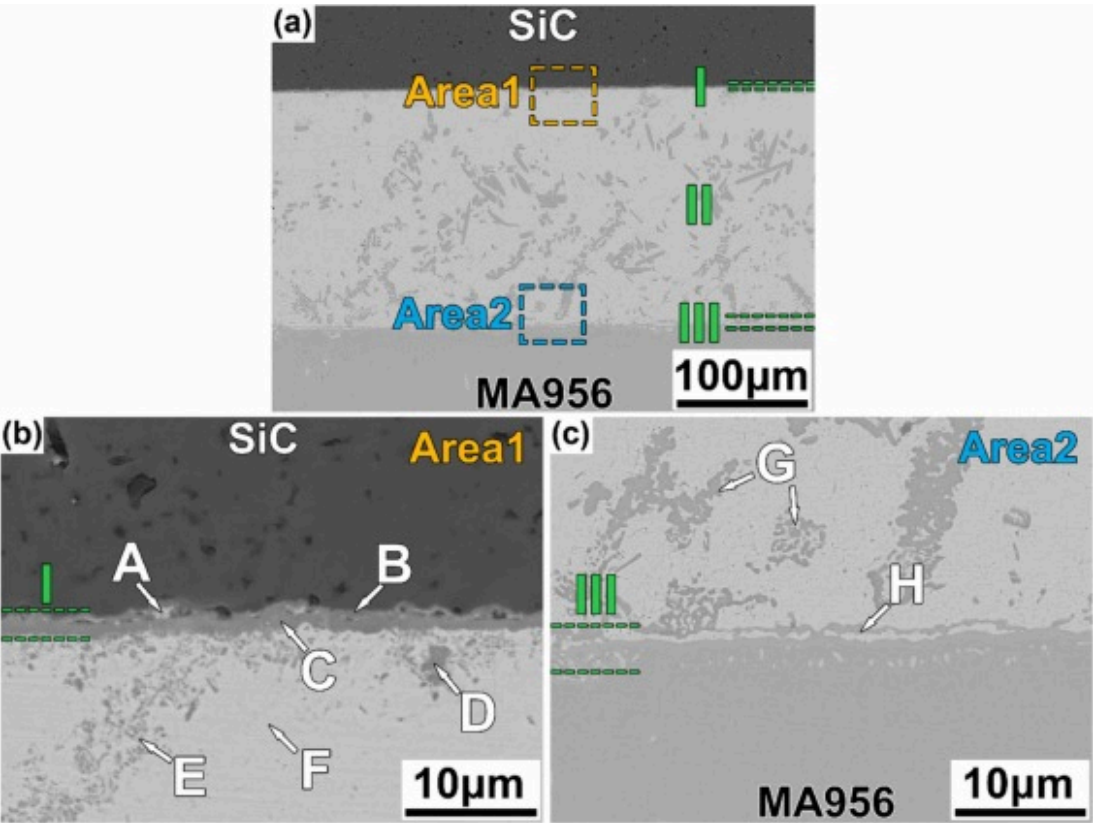
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Fig. 4. Cross-sectional interfacial microstructure and elemental mapping analysis of the SiC/Cu-Ti-5Si/MA956 ODS steel joint brazed at 960 °C for 10 min:(a) Overview morphology and elemental mapping of; (b) Si; (c) Fe; (d) Ti; (e) Cr; (f) Cu; (g) C.

Fig. 5 showed the interfacial morphology and local magnification analysis of SiC/Cu-Ti-5Si/MA956 ODS steel brazed joint ($T = 960$ °C, $t = 10$ min). Based on contrast and the distinct locations of reaction layers, the joint could be divided into three regions: Zone I (reaction layer adjacent to SiC), Zone II (central braze seam), and Zone III (reaction layer near the MA956 ODS steel). The decomposition products in Zone I were discontinuous and presented as fine granular particles. The reaction layer I was mainly composed of three phases, namely A, B, and C. Phase A was a discontinuous small particle phase, while phase B and phase C were continuous compound layer phases. The reaction layer II was primarily composed of four phases, namely D, E, F, and G. Both phase D and phase E were granular phases, and the color of phase D was darker than that of phase E. Phase F was a white matrix phase. The reaction layer III was mainly composed of phase H, which was a gray thin layer phase on the MA956 ODS steel side. EDS analysis was systematically performed on the constituent phases in Fig. 5 to determine their elemental composition and distribution patterns, and the elemental composition of each point was shown in Table 2. The gray-white phase A was mainly the decomposition product of SiC. According to the analysis of its energy spectrum composition results, it was composed of $\text{Cu}_{6.69}\text{Si}$ and graphite particles. Phase B was primarily a strip-shaped compound layer composed of two elements, Ti and C, and the composition ratio of the two elements was close to 1:1. Therefore, based on relevant literature [35], phase B was speculated to be the TiC phase. Phase C was mainly composed of 29.47 % Si and 58.18 % Ti. According to the Ti-Si binary phase diagram [20], it could be inferred that phase C was the Ti_5Si_3 phase. Phase D was a small particle phase on the SiC side, and its contrast and elemental composition were both close to those of phase C. Thus, it was speculated that phase D was the same as phase C, which was the Ti_5Si_3 phase. The lower stability of Ti_5Si_3 compared to TiC enabled preferential formation of TiC on the SiC surface, which subsequently developed into a continuous layer. This process creates a Si-depleted zone away from the SiC interface. Consequently, the slow nucleation of Ti_5Si_3 is suppressed by TiC, causing it to be pushed away from the interface and grow into the matrix. Phase E was a dissociating block phase, and its elemental composition was primarily Ti, Cu, and Si, with the ratio of the three elements approximately 1:1:1. Based on relevant literature [36], it was speculated to be the TiCuSi phase. The white phase F was the base phase of the brazing seam, and its main elemental composition was Cu. It was speculated to be the Cu(s,s) phase. Phase G was a gray compound on the MA956 ODS steel side, and its main elemental composition was 31.20 % Si, 34.41 % Cu, and 27.63 % Ti elements, which was close to the composition of phase E. It was speculated to also be the TiCuSi phase. Phase H was a thin layer phase on the base metal side of the MA956 ODS steel, and its main elemental composition was composed of

26.52 % Si, 33.42 % Ti, and 29.16 % Fe elements. According to its elemental composition and relevant literature [37], it was speculated to be the TiFeSi phase.



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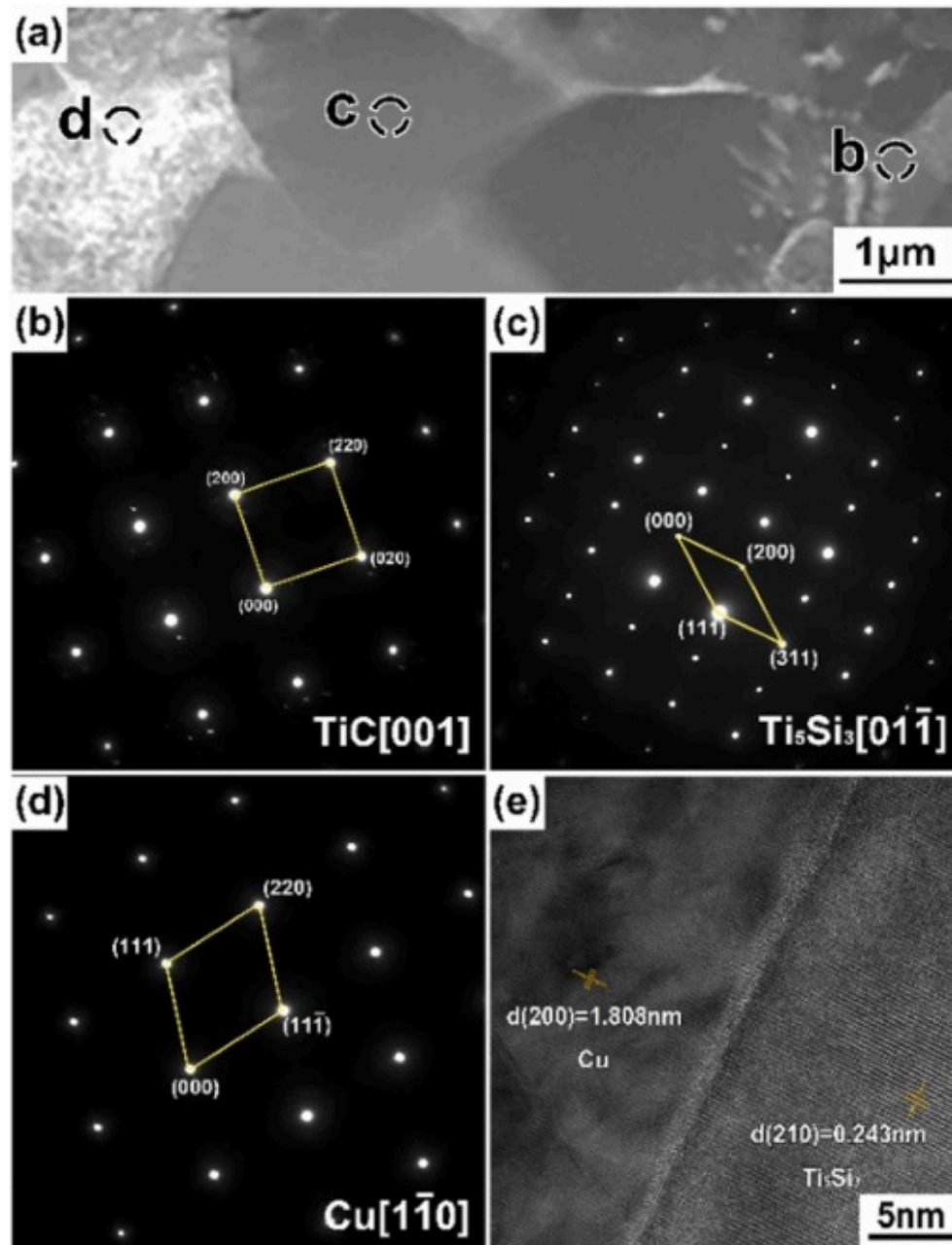
Fig. 5. Microstructural analysis of the typical interface in the SiC/Cu-Ti-5Si/MA956 ODS steel joint (T = 960 °C, t = 10 min):(a) Overall morphology; (b) Magnified view of the SiC-side interface; (c) Magnified view of the MA956-side interface.

Table 2. Chemical compositions of phases marked in Fig. 4 (at%).

Area	C	Si	Cu	Ti	Fe	Cr	Possible phase
A	19.20	11.43	66.37	1.32	0.83	0.85	Cu _{6.69} Si+C

Area	C	Si	Cu	Ti	Fe	Cr	Possible phase
B	39.36	4.96	2.19	50.45	1.29	1.75	TiC
C	3.26	29.47	4.88	58.18	2.73	1.48	Ti ₅ Si ₃
D	2.91	35.34	5.43	54.27	0.96	1.09	Ti ₅ Si ₃
E	4.23	29.26	33.22	31.19	0.65	1.45	TiCuSi
F	4.86	6.43	85.32	1.38	0.93	1.08	Cu(s,s)
G	1.71	31.20	34.41	27.63	4.23	0.82	TiCuSi
H	5.25	26.52	1.30	33.42	29.16	4.35	TiFeSi

To further determine the composition of these fine reaction phases, FIB technology was employed to extract samples from the joint shown in Fig. 5, followed by TEM analysis of the specimen's microstructure. Fig. 6 presented the morphology and elemental distribution of multiple reaction phases adjacent to the SiC ceramic. This characterization conclusively identified the reaction products of B, C, and F as TiC, Ti₅Si₃, and Cu(s,s) phases.



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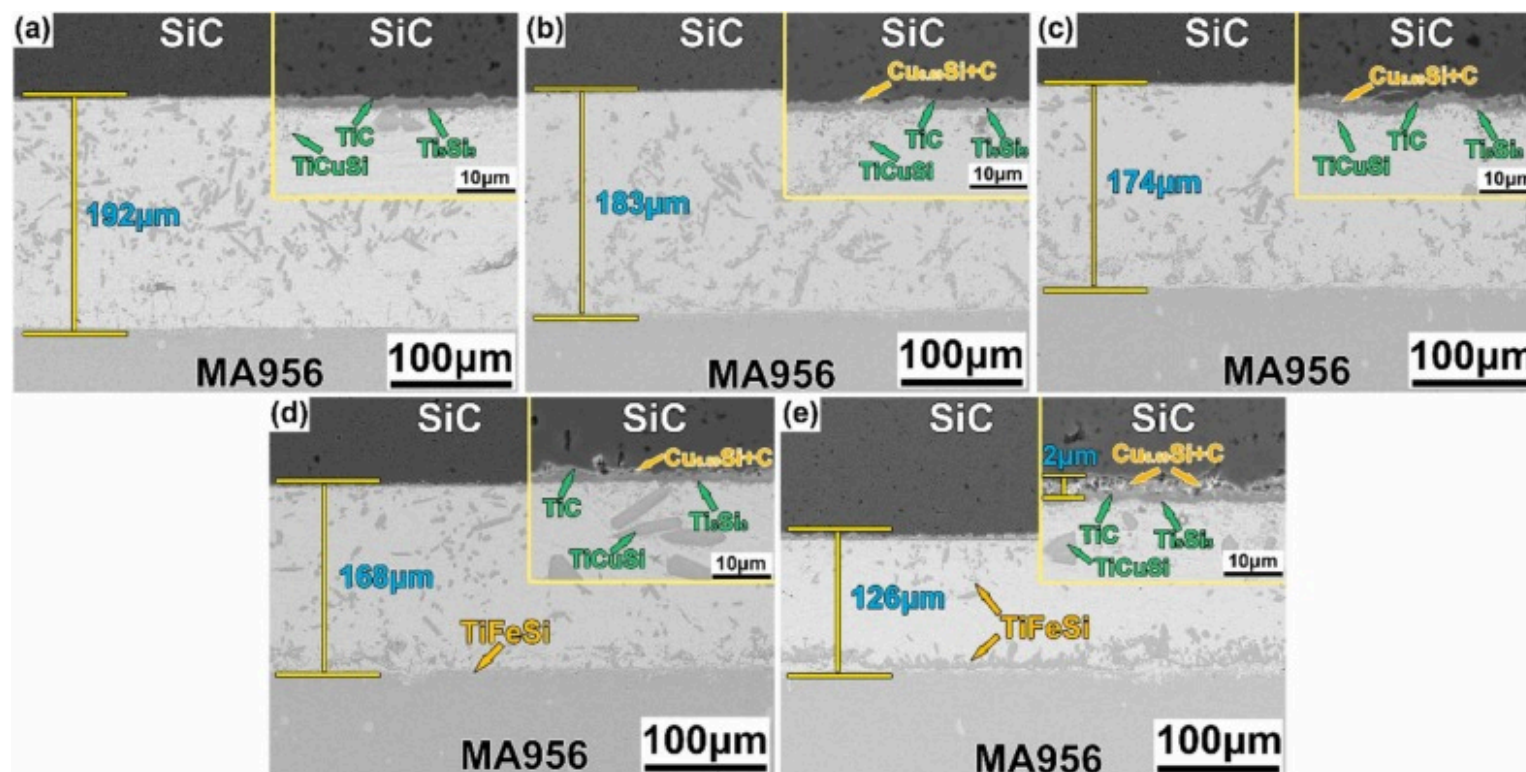
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Fig. 6. TEM analysis of the brazed joint: (a) The microstructure of the joint; (b-d) SAED and high-resolution images of phases marked in Fig. 5(a); (e) HRTEM image of the $\text{Ti}_5\text{Si}_3/\text{Cu}$ interface.

Based on the comprehensive analysis above, the interfacial microstructure of the joint fabricated via brazing SiC ceramic to MA956 (ODS) steel using a Cu-Ti-5Si filler alloy is characterized by SiC / $\text{Cu}_{6.69}\text{Si}$ + graphite / TiC / Ti_5Si_3 / TiCuSi + Ti_5Si_3 + Cu(s,s) / TiCuSi + Cu(s,s) / TiFeSi + Cu(s,s) / MA956.

3.3. Effect of brazing temperature and holding time on interfacial microstructure of SiC/MA956 joints

To investigate the influence of brazing temperature on the microstructure and mechanical properties of SiC/Cu-Ti-5Si/MA956 ODS steel joints, brazing experiments were conducted using a Cu-Ti-5Si filler alloy containing 5 wt% Si. The holding time was fixed at 10 min, while the brazing temperature varied across a range of 940–1020 °C. Fig. 7 showed the interfacial microstructures of brazed joints and magnified views of the SiC reaction interface.



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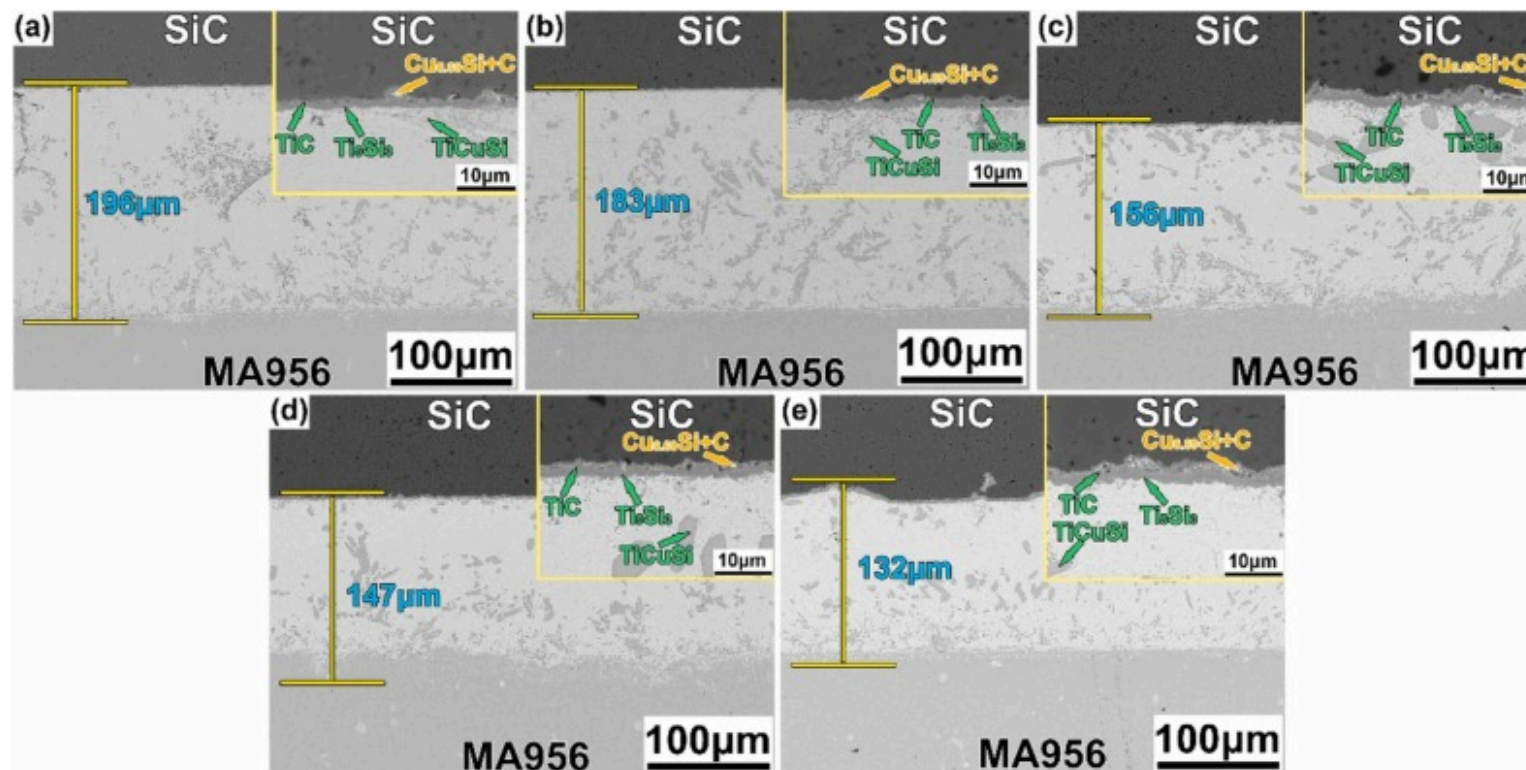
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Fig. 7. Interfacial microstructure of joints brazed at a distinct temperature for 10 min. (a) 940 °C; (b) 960 °C; (c) 980 °C; (d) 1000 °C; (e) 1020 °C.

As shown in Fig. 7, all joints brazed at different temperatures exhibited defect-free microstructures without voids or cracks. Measurements of brazing seam widths were conducted across specimens; as the brazing temperature increased, a pronounced narrowing trend in seam width was observed. At lower temperatures (940 °C), the limited duration of molten filler resulted in a seam width approximating the original filler thickness (~200 μm). The interfacial compounds primarily consisted of TiCuSi, TiC, and Ti₅Si₃. TiC and Ti₅Si₃ were predominantly located in the ~2 μm thick reaction layer on the SiC side. As the brazing temperature increased, the liquid filler metal's residence time extended, causing intensified outward flow and a gradual reduction in brazing seam width. Meanwhile, the thickness of the compound layer in Reaction Zone I showed a marginal increase

with negligible variation. At 1020 °C, the seam width reached a minimum of 126 μm , accompanied by accelerated SiC decomposition. Fig. 7(e) clearly presented a continuous 2 μm -thick decomposition layer at the SiC interface, however, compared to the 25.2 μm -thick SiC decomposition layer observed in Fig. 6(a), the current case was significantly mitigated. Notably, elevated brazing temperatures induced substantial dissolution and diffusion of Si/Fe elements from base material into the braze seam. This facilitated the transition of Ti-Fe-Si phases at the MA956 ODS steel interface from discontinuous granular particles to lamellar aggregates, which subsequently migrated toward the SiC side. This migration was attributed to the combined effects of enhanced thermal activation energy and a substantial reduction in braze seam width.

Fig. 8 showed the microstructures of brazed joints under different holding times at 960 °C. Notably, all joints exhibited robust bonding integrity under these conditions. At the brazing temperature (960 °C), SiC decomposition was significantly inhibited. No distinct SiC decomposition layer was detected on the SiC ceramic side at various holding times. The decomposition products of SiC ceramics were confined to Reaction Zone I as fine particles. It was demonstrated that the variation in holding time did not enhance the extent of Cu-induced decomposition reaction of SiC ceramics by the filler at this brazing temperature. Notably, at this brazing temperature, the TiFeSi intermetallic phase on the metallic side did not form a thicker compound layer despite prolonged holding time, remaining as discontinuous dispersions. Furthermore, TiFeSi phase diffusion was confined to the MA956 ODS steel side without detectable diffusion toward the SiC interface.



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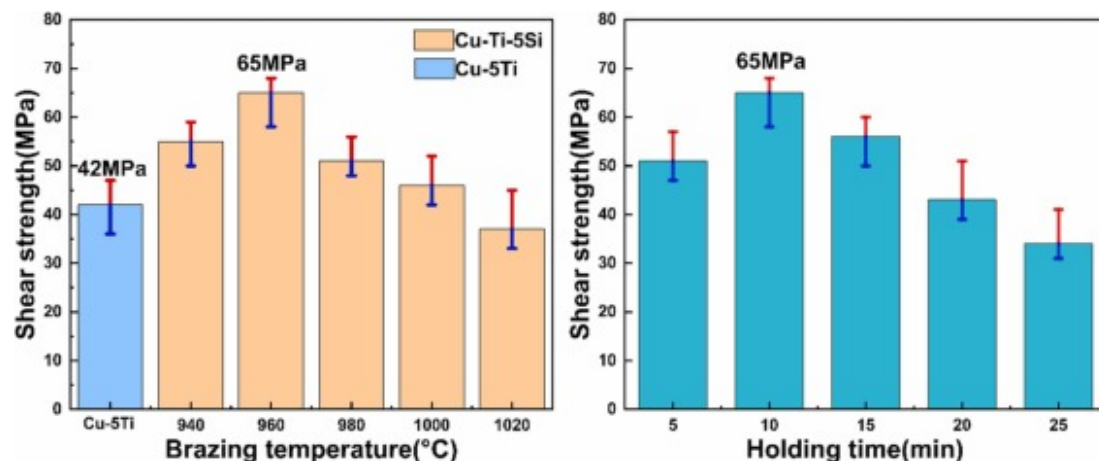
Fig. 8. Effect of holding time on the interfacial microstructure of SiC/Cu-Ti-5Si/MA956 ODS steel brazed joint ($T = 960\text{ }^{\circ}\text{C}$):(a) $t = 5\text{ min}$; (b) $t = 10\text{ min}$; (c) $t = 15\text{ min}$; (d) $t = 20\text{ min}$; (e) $t = 25\text{ min}$.

In conclusion, the Cu-Ti-5Si filler, compared to the Si-free Cu-5Ti filler, exhibited a melting point reduction of $101.4\text{ }^{\circ}\text{C}$. Concurrently, decomposition on the SiC side was significantly suppressed, limited decomposition at the SiC interface was detected exclusively at temperatures exceeding $1000\text{ }^{\circ}\text{C}$, resulting in a marked improvement in interfacial quality of the joint.

3.4. Effect of brazing temperature and holding time on mechanical properties of the SiC/MA956 joints

Fig. 9 was the variation in shear strength of SiC/MA956 ODS steel joints brazed using Cu-5Ti filler at $1060\text{ }^{\circ}\text{C}$ and Cu-Ti-5Si filler under different brazing temperatures and holding time. For the Cu-Ti-5Si filler, as the brazing temperature and holding time rose,

the shear strength of the joint increased and then decreased. Su et al. demonstrated that the SiC/MA956 ODS steel joint brazed with Si-free Cu-5Ti filler achieved a maximum shear strength of 42 MPa under a brazing condition of 1060 °C for 10 min [34]. In contrast, this made Cu-Ti-5Si filler exhibited a significantly higher maximum shear strength of 65 MPa at 960 °C under the same holding time, representing a 54.8 % strength improvement.



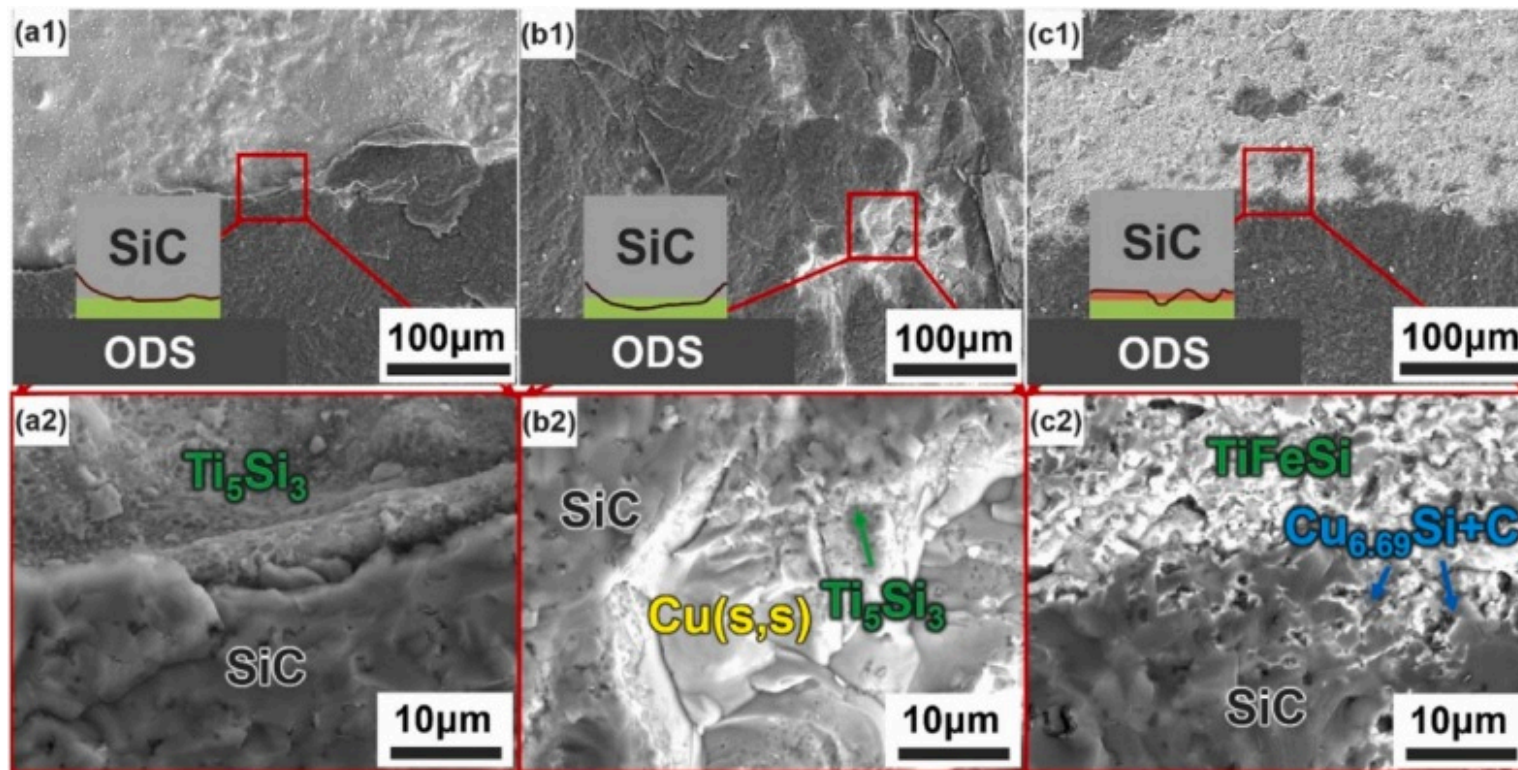
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Fig. 9. Influence of process parameters on shear strength of brazed joints of steel.

Fig. 10 presented the fracture locations and morphologies of SiC/MA956 ODS steel joints brazed for 10 min at different temperatures. Brazing temperature significantly affected the fracture behavior of the joint. The fracture morphology consistently exhibited a brittle mode under different temperatures. At a lower brazing temperature of 940 °C, fracture occurred at the interface between SiC ceramic and filler. The fracture surface was dominated by SiC substrate and Ti_5Si_3 phases. This phenomenon arose from the shorter residence time of liquid filler metal at lower brazing temperature, leading to insufficient interfacial reactions between the base metal and filler. Consequently, a robust bonding interface failed to form, creating a weak zone within the joint. Under external loading, fracture preferentially initiated at this underdeveloped interfacial region. At a brazing temperature of 960 °C, the joint fracture path traversed Reaction Zone II. The Cu(s,s) phase was detected in the fracture morphology, and fracture propagation occurred through both Reaction Zone II and the SiC-side Reaction Zone I. The mechanical properties of the brazed joint reached a maximum shear strength of 65 MPa at this temperature. At a brazing temperature of

1020 °C, a SiC decomposition layer and brittle TiFeSi phase were detected on the joint fracture surface. This indicated that the fracture site shifted to the SiC ceramic decomposition zone at this temperature. At elevated brazing temperatures, the Cu-induced decomposition of SiC ceramics intensified, thickening the interfacial decomposition layer of Cu-Si compounds and graphite. Meanwhile, the pronounced CTE mismatch between SiC ceramics and MA956 ODS steel generated significant residual stress at the interface. Together, these factors established the SiC decomposition zone as a weak region, leading to fracture propagation within this area.

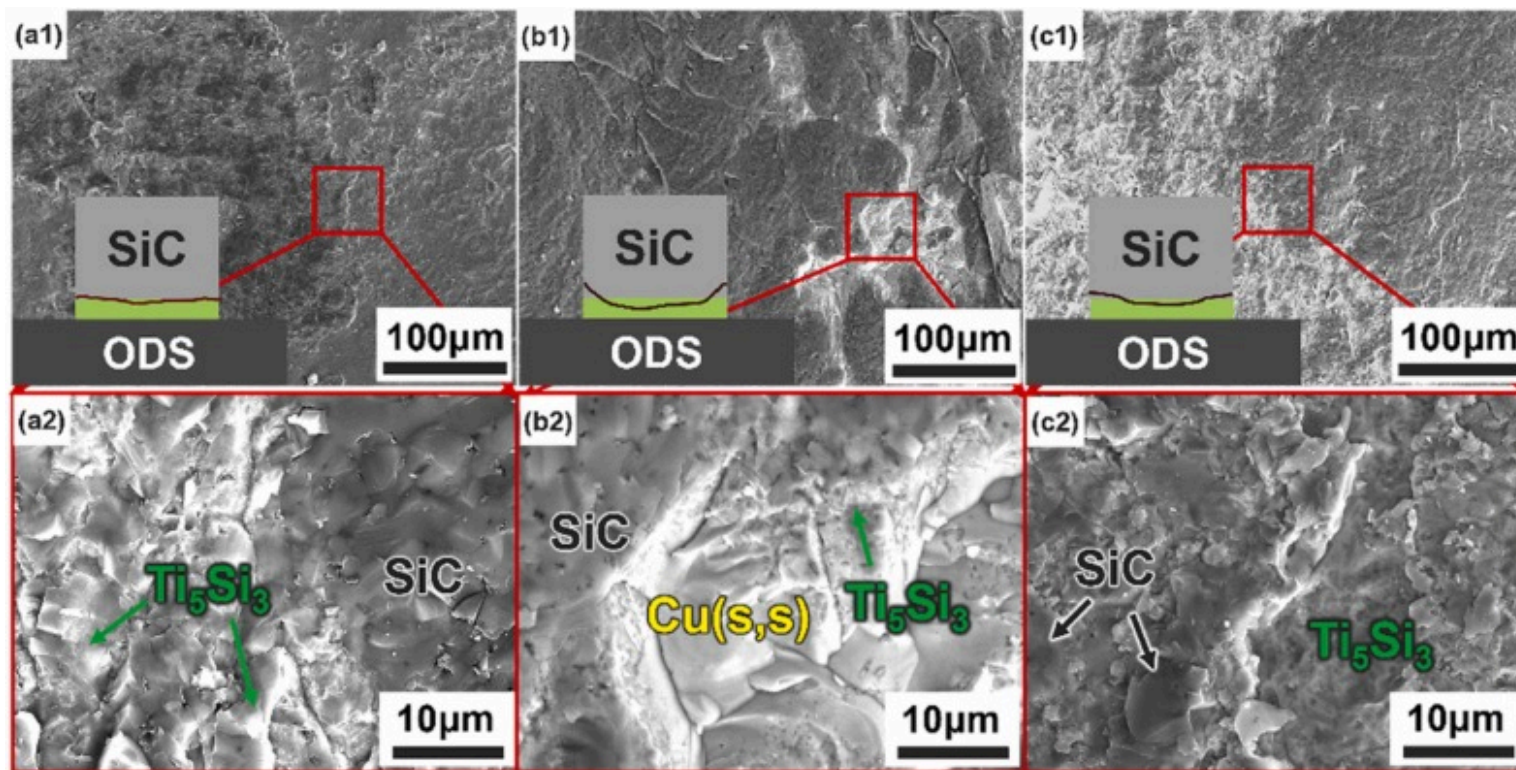


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Fig. 10. Fracture morphologies of the brazed joints of SiC/Cu-Ti-Si/MA956 ODS steel at different brazing temperatures ($t = 10$ min): (a1, a2) 940 °C; (b1, b2) 960 °C; (c1, c2) 1020 °C.

Fig. 11 revealed that holding time significantly affected both fracture location and morphology of brazed joints. All fracture surfaces exhibited brittle fracture characteristics. At 5 min holding time, fracture occurred at the SiC/filler interface, with the fracture surface dominated by SiC substrate and Ti_5Si_3 phases. This occurred because the short holding time limited the duration of liquid filler availability, leading to inadequate interfacial reaction between the substrate and filler. Consequently, weak interfacial bonding developed, making this region the preferred crack initiation site under applied stress. When holding time extended to 10 min, fracture propagated through Reaction Zone II, and $Cu(s,s)$ was detected on the fracture surface. This condition corresponded to peak joint strength. Further extension to 25 min led to fracture primarily within Reaction Zone I, evidenced by Ti_5Si_3 and limited SiC particles. Here, Reaction Zone I (composed of brittle Ti_5Si_3 and TiC) thickened to $\sim 6\ \mu m$. Accumulation of these brittle phases embrittled the joint, causing fracture initiation in Reaction Zone I under stress.



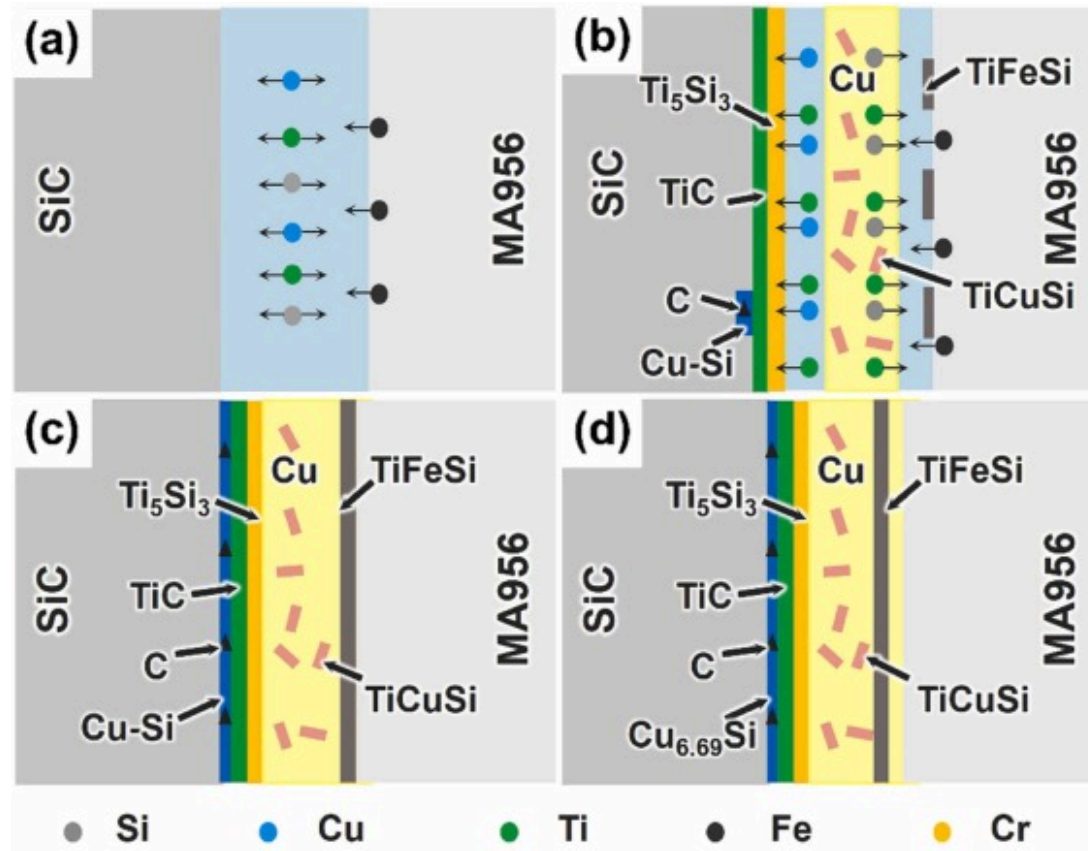
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Fig. 11. Fracture morphologies of brazed joints of SiC/Cu-Ti-Si/MA956 ODS steel under different holding times ($T = 960\text{ }^{\circ}\text{C}$):(a1, a2) 5 min; (b1, b2) 10 min; (c1, c2) 25 min.

4. Discussion

To clarify the formation mechanism of the Cu-Ti-5Si brazed joint, a schematic diagram showing the joint formation process was presented in Fig. 12. The formation process of the SiC/Cu-Ti-5Si/MA956 ODS steel joint could be divided into the following stages:



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Fig. 12. Schematic diagram of the formation process of the joint between SiC and MA956 ODS steel brazed with Cu-Ti-5Si filler metal:(a) Melting stage of the filler metal; (b) Initial reaction stage; (c) Stage of thickening of the reaction layer and stripping of

the TiFeSi phase; (d) Stage of precipitation of the residual liquid phase.

- (1) **Liquid Phase Formation Stage.** Prior to reaching the melting point of the Cu-Ti-5Si filler, the joint remained in its initial state. As the furnace temperature gradually increased, when the brazing filler reached its melting point, it initiated melting and formed a liquid phase. Driven by chemical potential gradients, Ti atoms from the liquid filler diffused toward both base metals (SiC and MA956 ODS steel), while Cu atoms preferentially diffused toward the SiC side. Concurrently, Si atoms were transported with the liquid phase toward the steel side, and Fe atoms from the MA956 ODS steel diffused into the braze seam.
- (2) **Initial Reaction Stage.** The SiC ceramic first reacted with Ti atoms from the filler alloy, forming TiC and Ti_5Si_3 phases, as described by Fig. 12(b). Ti combined with carbon decomposed from SiC to generate a high-melting-point TiC layer (3140 °C), as the primary reaction product (located at the SiC side). When the concentrations of Ti and Si in the liquid phase were sufficiently high, the Ti_5Si_3 phase (melting point: 2130 °C) formed between the TiC layer and the braze seam. Concurrently, the Cu-induced decomposition of SiC continued; however, its degree of progression was governed by the Si content in the Cu-rich liquid and Cu-Si liquid phases. The addition of Si to the brazing filler increased the Si content in the Cu-Si liquid phase, thereby significantly suppressing the decomposition reaction. At the MA956 ODS steel interface, Si and Ti dissolved in the molten filler diffused toward the steel substrate. Within the Si-Ti-Fe ternary system, the TiFeSi phase formed preferentially because of its thermodynamic stability, and it localized on the MA956 ODS steel side during brazing.
- (3) **Reaction Layer Thickening and TiFeSi Delamination Stage.** As the brazing temperature increased or the holding time prolonged, the existence duration of the liquid phase was prolonged, and the intermetallic compound layer was continuously thickened. On the SiC side, elevated temperatures raised the Si saturation concentration in the Cu-Si liquid phase, intensified Cu-induced decomposition of SiC. This resulted in the transformation of SiC decomposition products from discontinuous granular structures to continuous layered configurations, but the overall extent of decomposition was substantially suppressed. On the MA956 ODS steel side, driven by the chemical potential gradient, Fe atoms strongly diffused toward the braze seam center and SiC side, leading to detachment of the TiFeSi phase in the microstructure.

- (4) **Residual Liquid Phase Precipitation Stage.** During the subsequent cooling process, the residual liquid phase within the joint gradually solidifies and precipitates. When the furnace temperature decreases to the melting point of $\text{Cu}_{6.69}\text{Si}$, the remaining Cu-Si liquid phase begins to solidify and precipitate, ultimately forming a $\text{Cu}_{6.69}\text{Si}$ + graphite decomposition layer adjacent to the SiC ceramic.

5. Conclusion

- (1) The typical interfacial configuration of the SiC/Cu-Ti-5Si/MA956 ODS steel brazed joint is characterized as: SiC / $\text{Cu}_{6.69}\text{Si}$ + graphite / TiC / Ti_5Si_3 / TiCuSi + Ti_5Si_3 + Cu(s,s) / TiCuSi + Cu(s,s) / TiFeSi + Cu(s,s) / MA956.
- (2) The decomposition of SiC ceramics in joints brazed with the Cu-Ti-5Si filler is significantly suppressed. Continuous layered decomposition products of SiC are observed only at the maximum brazing temperature of 1020 °C, while variations in holding time exhibit negligible effects on SiC decomposition. Both increasing brazing temperature and prolonging holding time led to a marked reduction in braze seam width. The TiFeSi phase within the seam gradually aggregates into layered structures with rising temperature but remains discontinuous under extended holding times.
- (3) The shear strength of the joints initially increases and subsequently decreases with elevated brazing temperatures or prolonged holding times. Under optimal parameters (960 °C, 10 min), the fracture path propagates through the ductile Cu(s,s) phase, achieving a peak shear strength of 65 MPa, the joint strength exhibited a 54.8 % enhancement compared to those fabricated with Cu-5Ti fillers.

CRediT authorship contribution statement

X.G. Song: Funding acquisition and Writing – review & editing. **X.P. Zhao:** Writing – original draft and Writing – review & editing. **S. Su:** Writing – original draft and Writing – review & editing. **W.L. Zhou:** Writing – review & editing. **W. Fu:** Funding acquisition. **Y.Y. Song:** Writing – review & editing. **F. Long:** Writing – review & editing. **J. Qin:** Writing – review & editing. **S.P. Hu:** Funding acquisition and Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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